24. Design a C Program to Demonstrate UNIX System Calls for File Management

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Code - Part 1 (Lines 1-34) -- File Creation, Opening & Writing

```c
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <unistd.h>
4  #include <fcntl.h>
5  #include <sys/types.h>
6  #include <sys/stat.h>
7
8  int main() {
9      int fd;
10     char buffer[100];
11
12     // Creating a file
13     fd = creat("sample.txt", S_IRWXU);
14     if (fd == -1) {
15         perror("create");
16         exit(1);
17     } else {
18         printf("File 'sample.txt' created successfully.\n");
19         close(fd);
20     }
21
22     // Opening file for writing
23     fd = open("sample.txt", O_WRONLY | O_APPEND);
24     if (fd == -1) {
25         perror("open");
26         exit(1);
27     } else {
28         printf("File 'sample.txt' opened for writing.\n");
29     }
30
31     // Writing data
32     write(fd, "Hello, World!\n", 14);
33     printf("Data written to 'sample.txt'.\n");
34     close(fd);
```

Code - Part 2 (Lines 35-66) -- File Reading, Deletion & Return

```
35
36      // Opening file for reading
37      fd = open("sample.txt", O_RDONLY);
38      if (fd == -1) {
39          perror("open");
40          exit(1);
41      } else {
42          printf("File 'sample.txt' opened for reading.\n");
43      }
44
45      // Reading data
46      int bytesRead = read(fd, buffer, sizeof(buffer));
47      if (bytesRead == -1) {
48          perror("read");
49          exit(1);
50      } else {
51          printf("Data read from 'sample.txt':\n");
52          write(STDOUT_FILENO, buffer, bytesRead);
53      }
54
55      close(fd);
56
57      // Deleting file
58      if (remove("sample.txt") == -1) {
59          perror("remove");
60          exit(1);
61      } else {
62          printf("File 'sample.txt' deleted.\n");
63      }
64
65      return 0;
66  }
```

Output

```
File 'sample.txt' created successfully.
File 'sample.txt' opened for writing.
Data written to 'sample.txt'.
File 'sample.txt' opened for reading.
Data read from 'sample.txt':
Hello, World!
File 'sample.txt' deleted.


...Program finished with exit code 0
Press ENTER to exit console.
```